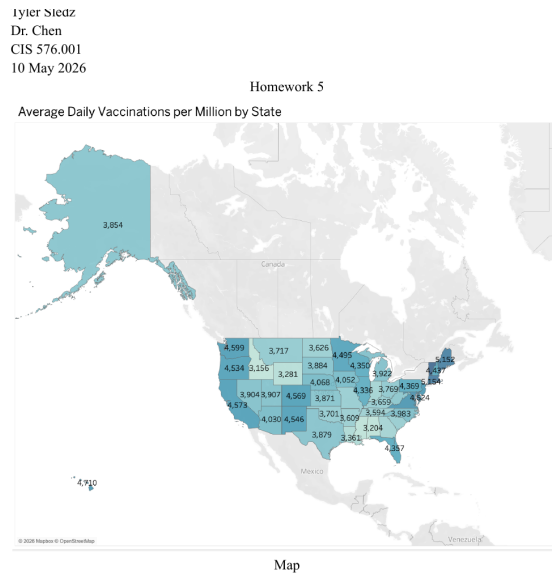


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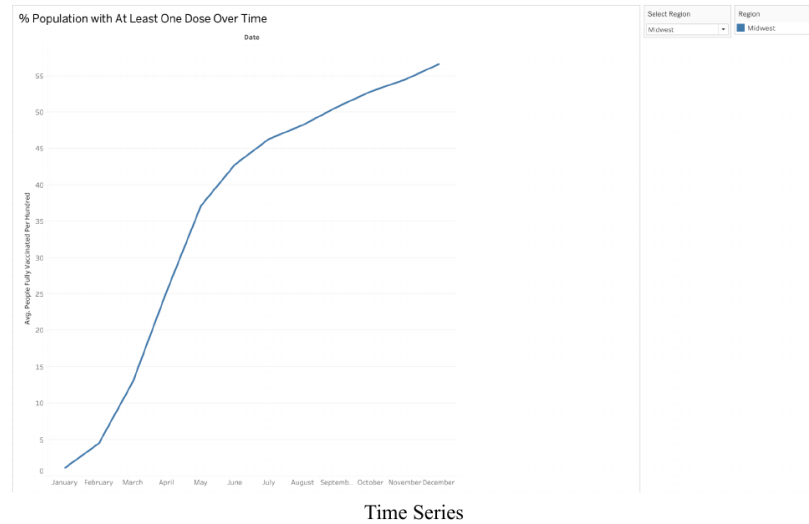
Homework 5: U.S. COVID-19 Vaccination Dashboard

1. Data Map: Average Daily Vaccinations per Million by State



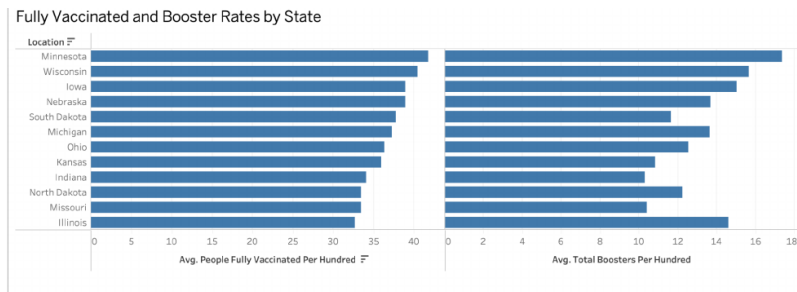
Key Insights: The choropleth map reveals meaningful geographic variation in average daily vaccination rates across U.S. states in 2021. Northeastern states such as Maine and Connecticut recorded the highest rates (above 5,000 per million), reflecting early and aggressive vaccine rollout programs. Southern and some Midwestern states — including Mississippi and Missouri — lagged behind with rates closer to 3,200 per million. Alaska stands out among non-contiguous states with a relatively high vaccination rate of 3,854 per million. Overall, the map suggests a coastal vs. interior divide in vaccination pace.

2. Time Series: % Population with At Least One Dose Over Time



Key Insights: The time series shows a consistent S-curve pattern across all four U.S. regions, with rapid vaccination uptake beginning in March 2021 and plateauing by mid-summer. The Northeast region (orange line) led all other regions throughout the year, reaching over 70% of its population with at least one dose by December — nearly 15 percentage points ahead of the South (red line), which consistently trailed. The Midwest and West followed similar trajectories, converging around 55-57% by year end. The mid-summer slowdown visible across all regions reflects nationwide vaccine hesitancy that emerged after the initial rollout surge.

3. Bar Chart: Fully Vaccinated and Booster Rates by State



Bar Chart

Key Insights: The side-by-side bar chart compares average fully vaccinated rates (left) and booster rates (right) by state, filtered by region. Among Midwest states shown, Minnesota and Wisconsin led in both fully vaccinated rates (approximately 40 per hundred) and booster adoption (16-18 per hundred). Missouri and Illinois consistently ranked at the bottom of both metrics within the region. Notably, states with high full vaccination rates did not always lead in boosters — suggesting that booster uptake may be driven by different demographic or policy factors than initial vaccination completion.

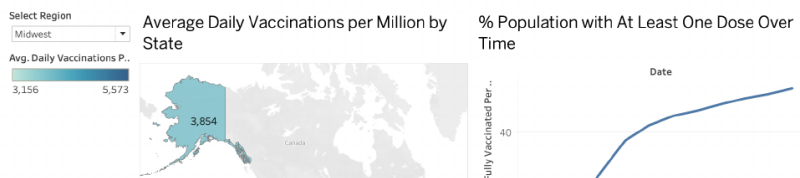
4. Data Table: Vaccination Summary Table by State

Vaccination Summary Table by State

Location	Vaccinated Per Hundred	Avg. Distributed Per Hundred	Avg. Total Boosters Per Hundred
Illinois	33	92	15
Indiana	34	93	10
Iowa	39	101	15
Kansas	36	100	11
Michigan	37	103	14
Minnesota	42	105	17
Missouri	33	95	10
Nebraska	39	101	14
North Dako...	33	91	12
Ohio	36	97	13
South Dako...	38	103	12
Wisconsin	40	98	16

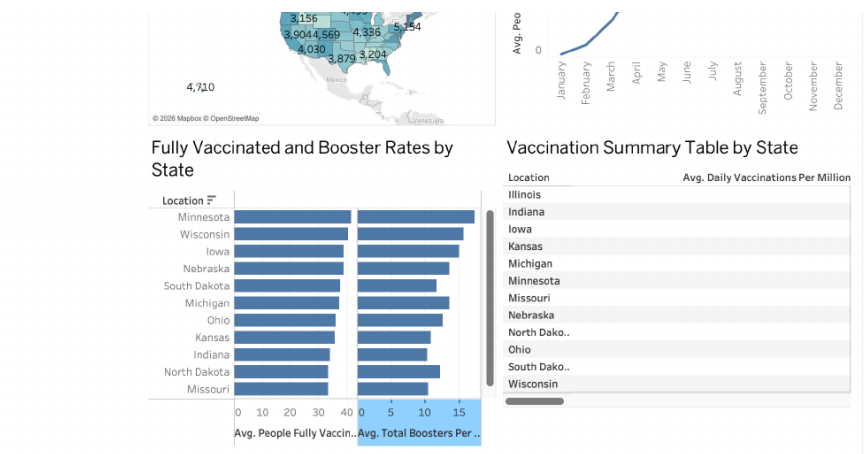
Data Table

U.S. COVID-19 Vaccination Dashboard 2021



Key Insights: The data table provides a precise numerical summary of five key vaccination metrics by state, filterable by region. For the Midwest region displayed, Minnesota consistently ranks highest across vaccinated per hundred (42), distributed per hundred (105), and boosters per hundred (17), indicating both strong supply and strong demand. Missouri and Illinois show the lowest values across most metrics, with vaccinated per hundred as low as 33. The distributed per hundred column reveals that all Midwest states received more than 90 doses distributed per 100 residents, suggesting supply was not the primary limiting factor — rather, uptake and hesitancy drove differences in outcomes.

5. Dashboard Overview



Dashboard

Key Insights: The interactive dashboard integrates all four visualizations into a unified view with a Select Region parameter that simultaneously filters the time series, bar chart, and data table to the chosen region while the map retains a national view for geographic context. When set to Midwest (shown), the dashboard tells a cohesive regional story: Midwestern states achieved moderate vaccination rates driven primarily by Minnesota and Wisconsin, with visible mid-year slowdowns in the time series and meaningful variation in booster uptake across states. The dashboard design follows effective visualization principles with a clear title, consistent chart titles, and a clean 2x2 layout.